Def. Let $f_n : X \to [-\infty, \infty]$ $(n \in \mathbb{N})$ be functions.
Define for each $x \in X$,

$$\left(\sup_{n\in\mathbb{N}} f_n\right)(x) \overset{\text{def}}{=} \sup_{n\in\mathbb{N}} f_n(x) \quad \text{and} \quad \left(\limsup_{n\to\infty} f_n\right)(x) = \limsup_{n\to\infty} f_n(x).$$

Lemma. Let $f : X \to [-\infty, \infty]$ with X is equipped a σ-algebra $\mathcal{M}$.
If $f^{-1}[(a, \infty]] \in \mathcal{M}$ for every $a \in \mathbb{R}$, then f is measurable.

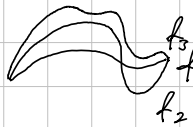
Proof. Let $\Omega := \{E \subseteq [-\infty, \infty] \mid f^{-1}[E] \in \mathcal{M}\}$. Given $a \in \mathbb{R}$, let (a_n) be a sequence such that $a_n < a$ for all $n \in \mathbb{N}$ and $a_n \to a$ as $n \to \infty$.
(Concretely, we can choose $a_n := a - 1/n$).
By the assumption, for each n, $f^{-1}[(a_n, \infty]] \in \mathcal{M}$ so that $(a_n, \infty] \in \Omega$.
Therefore,

$$[-\infty, a) = \bigcup_{n=1}^{\infty} [-\infty, a_n] = \bigcup_{n=1}^{\infty} (a_n, \infty]^c \in \Omega$$

because Ω is also σ-algebra, and thus $f^{-1}[[-\infty, a)] \in \mathcal{M}$.
Moreover, given $a, b \in \mathbb{R}$, $(a, b) = [-\infty, b) \cap (a, \infty]$ so that

$$f^{-1}[(a,b)] = f^{-1}[[-\infty, b)] \cap f^{-1}[(a, \infty]] = \left(f^{-1}[[-\infty, b)]^c \cup f^{-1}[(a, \infty]]^c\right)^c \in \mathcal{M}. \quad \square$$

Thm. Let $f_n : X \to [-\infty, \infty]$ $(n \in \mathbb{N})$ be measurable functions. Then,
$\sup_{n\in\mathbb{N}} f_n$ and $\limsup_{n\in\mathbb{N}} f_n$ are measurable.



Proof. Given $a \in \mathbb{R}$,

$$\left(\sup_{n\in\mathbb{N}} f_n\right)^{-1}[(a, \infty]] = \{x \in X \mid \sup_{n\in\mathbb{N}} f_n(x) \in (a, \infty]\} = \bigcup_{n\in\mathbb{N}} f_n^{-1}[(a, \infty]]$$

because given $x \in X$, $a < \sup_{n\in\mathbb{N}} f_n(x)$ implies $a < f_N(x)$ for some $N \in \mathbb{N}$.
Conversely, $a < f_n(x)$ for some $n \in \mathbb{N}$, then $a < f_n(x) \le \sup_{n\in\mathbb{N}} f_n(x)$, so
by the lemma above, the $\sup f_n$ is measurable.
To show $\limsup f_n$ is measurable, observe

$$\limsup_{n\to\infty} f_n = \inf_{n\in\mathbb{N}} \left\{\sup_{k\ge n} f_n\right\}, \text{ so proved.} \quad \square$$

[Simple function.
Def. Let X be a measurable space.
A complex-valued map $S: X \to \mathbb{C}$ is called a simple function if:

$$\text{range}(S) \subset \mathbb{C} \text{ is finite}$$

Let $\text{range}(S) = \{a_1, \cdots, a_n\}$. For each $1 \le i \le n$,
$A_i = S^{-1}[\{a_i\}] = \{x \in X \mid S(x) = a_i\}$ and $\chi_i \overset{\text{def}}{=} \chi_{A_i}$.

Then, $S = \sum_{i=1}^{n} a_i \cdot \chi_i$. Particularly,
[S is measurable if and only if for all $1 \le i \le n$, A_i is measurable.

Proof. Suppose $S: X \to \mathbb{C}$ is measurable.
Given $a_i \in \text{range}(S)$, there is an open set $\mathcal{U}_i$ such that

$$\mathcal{U}_i \cap \text{range}(S) = \{a_i\}, \text{ since the range is finite.}$$

Now, $S^{-1}[\mathcal{U}_i] = S^{-1}[\{a_i\}] = A_i$ is measurable.
Conversely, suppose A_i is measurable for each $1 \le i \le n$.
Given an open set $\mathcal{U} \subseteq \mathbb{C}$,

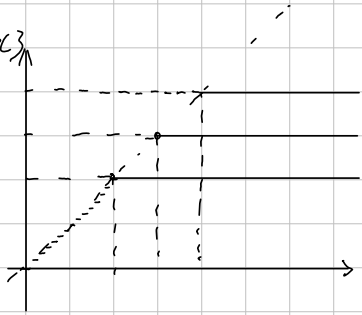
$$S^{-1}[\mathcal{U}] = \bigcup_{\{i : a_i \in \mathcal{U}\}} A_i$$

is a finite union of measurable set,
So measurable, we are done. ☑

Thm. Let $f: X \to [0, \infty]$ be measurable.
Then, there exist simple measurable functions s_n on X such that
$0 \le s_1 \le s_2 \le \cdots \le f$ and $s_n \to f$ pointwise on X.

Proof. For each $n \in \mathbb{N}$, define for $t \in \mathbb{R}$,

$$k_n(t) = \lceil t \cdot 2^n \rceil \quad \text{where } \lceil x \rceil := \max\{n \in \mathbb{Z} \mid n \le x\}$$



Then, $k_n(t) \le t \cdot 2^n < k_n(t) + 1$ and so that

$$k_n(t) \cdot 2^{-n} \le t < (k_n(t) + 1) \cdot 2^{-n}.$$

Define $\varphi_n(t) := \begin{cases} k_n(t) \cdot 2^{-n} & \text{if } 0 \le t < n \\ n & \text{if } n \le t \le \infty \end{cases}$

Then, φ_n is Borel map for each $n \in \mathbb{N}$, because every preimage of φ_n is countable. Moreover,

for each $0 \le t \le n$, $t - 2^{-n} < \varphi_n(t) \le t$

and $0 \le \varphi_1 \le \varphi_2 \le \cdots \le t$ and $\varphi_n(t) \to t$ pointwise on $[0, \infty]$
Now, for each $n \in \mathbb{N}$, take

$s_n := \varphi_n \circ f$. Then, $0 \le s_1 \le s_2 \le \cdots \le f$ and $s_n \to f$ pointwise. □

Note: if such $\{s_n\}$ exists, then f is measurable, since

$$f = \sup_{n \in \mathbb{N}} f_n$$

Therefore,

$f: X \to [0, \infty]$ is measurable if and only if $\exists \{s_n\}$ s.t $s_n \nearrow f$.